Results--Periodic Properties Lab Name _____

Part 1—Qualitative Identification Scheme

1. Based on your experimental observations, what can you conclude about the relative solubilities of the halogens and halides in mineral oil? Explain the preference for one solvent or the other in terms of polarity.

Part 2—Halogens as Oxidizing Agents

2. Fill in table below.

	Cl_2		Br_2		I_2	
	Rxn? (Yes/No)	Color of oil layer	Rxn? (Yes/No)	Color of oil layer	Rxn? (Yes/No)	Color of oil layer
Cl^-						
Br^-						
I^-						

3. Arrange the halogens in order of increasing strengths as oxidizing agents.

_____ < _____ < _____

Briefly explain how your experimental observations led you to this ordering.

	Cl_2		Br_2		I_2	
	Rxn? (Yes/No)	observations	Rxn? (Yes/No)	observations	Rxn? (Yes/No)	observations
Copper (Cu)						

4. For each reaction which occurred in **Part 2**, write a balanced net ionic chemical redox reaction. In addition, clearly label the species being oxidized and reduced in each reaction.

Part 3—Halides as Reducing Agents

5. Show the table which summarizes the experiments you designed to determine the relative strengths of halides as reducing agents along with your observations for each experiment.

6. What do you conclude about the relative reducing strengths of the halides? Support your statements by referring to experimental results.

7. For each reaction which occurred in **Part 3**, write a balanced net ionic chemical redox reaction. In addition, clearly label the species being oxidized and reduced in each reaction.
   Note: Any reaction involving potassium permanganate in this lab will produce manganese dioxide (MnO_2) as one of the products. Any reaction involving copper sulfate in this lab will produce copper (I) halide (CuX) as one of the products.